Muscle Symptoms Define a Stable Physiological Dysregulation Cluster in ME/CFS: Sex- and Menopause-Related Differences in Symptom Expression

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Muscle Symptoms Define a Stable Physiological Dysregulation Cluster in ME/CFS: Sex- and Menopause-Related Differences in Symptom Expression

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Abstract

Background: Muscle-related symptoms are among the most disabling manifestations of myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS), yet their position within the broader symptom architecture of the disease remains unclear. We hypothesized that muscle symptoms function as bridge symptoms linking multiple dysregulated physiological systems and that their symptom associations differ according to sex and menopausal status.

Methods: Cross-sectional data from 745 individuals with physician-diagnosed ME/CFS enrolled in the APAV-ME/CFS registry were analyzed. Muscle problems served as the primary outcome variable. Associations with 14 symptom domains were examined using multivariable logistic regression. Pearson and tetrachoric correlations, exploratory factor analysis, and structural equation modeling (SEM) were used to identify latent symptom structures. Analyses were stratified by sex, menopausal status, and disease duration.

Results: In the full multivariable model, breathing problems ($\beta = 0.96$, $p = 0.001$), flu-like symptoms ($\beta = 0.65$, $p = 0.020$), and temperature-regulation disorders ($\beta = 0.63$, $p = 0.030$) independently predicted muscle problems. A reduced physiological model additionally identified cardiovascular symptoms ($\beta = 0.65$, $p = 0.017$) and visual disturbances ($\beta = 0.52$, $p = 0.046$) as significant predictors. Tetrachoric correlations demonstrated substantial latent associations between muscle problems and breathing difficulties ($\rho = 0.54$), cardiovascular symptoms ($\rho = 0.50$), temperature-regulation disorders ($\rho = 0.50$), visual disturbances ($\rho = 0.41$), and flu-like symptoms ($\rho = 0.39$). Factor analysis and SEM supported a single latent physiological dysregulation factor with good model fit (RMSEA = 0.046, CFI = 0.972, TLI = 0.953, SRMR = 0.026). Sex-stratified analyses revealed distinct symptom architectures. In women, all five physiological symptoms independently predicted muscle problems, whereas in men only breathing problems and temperature-regulation disorders remained significant. Women classified as premenopausal showed strongest associations with cardiovascular symptoms, visual disturbances, and temperature dysregulation, while women classified as postmenopausal displayed a distinct symptom pattern characterized by breathing difficulties and flu-like symptoms. The latent physiological factor remained stable across disease-duration groups. Gastrointestinal complaints loaded substantially on the same latent factor despite lacking independent predictive value in regression analyses, whereas urogenital symptoms did not.

Conclusions:

Muscle symptoms occupy a central position within a stable physiological dysregulation cluster in ME/CFS encompassing respiratory, cardiovascular, thermoregulatory, visual, flu-like, and gastrointestinal manifestations. These findings provide evidence for a stable physiological symptom architecture that may improve clinical phenotyping and facilitate future biomarker-guided stratification The identified latent physiological factor remained stable across disease duration but showed distinct clinical expressions according to sex and menopausal status. Women classified as premenopausal were characterized primarily by cardiovascular, visual, and thermoregulatory symptom associations, whereas women classified as postmenopausal and men showed greater prominence of breathing-related symptoms. The persistence of flu-like symptoms in women classified as postmenopausal suggests that menopause modifies symptom expression without abolishing the underlying physiological dysregulation. More broadly, the findings are compatible with the hypothesis that this physiological dysregulation factor reflects a chronic PEM-associated multisystem response involving interacting autonomic, vascular, immunological, and metabolic pathways. Incorporating sex and menopausal status into ME/CFS phenotyping may improve patient stratification and facilitate the development of mechanism-based therapeutic approaches.

Keywords:

- Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS)
- Post-exertional malaise
- Muscle symptoms
- Physiological dysregulation
- Sex differences

Introduction

Muscle-related symptoms are among the most frequent and disabling manifestations of myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS), yet their pathophysiological basis remains incompletely understood. Patients commonly report exertional muscle pain, very early fatigability, muscle twitching and cramps, tremor, heaviness—even to the point of a complete inability to perform movements—as well as severely delayed recovery following even minimal physical activity [1-3]. These symptoms contribute substantially to functional impairment and are closely linked to post-exertional malaise (PEM), the cardinal feature of ME/CFS. Previous studies have consistently demonstrated that musculoskeletal pain and exercise-induced symptom exacerbation represent major determinants of disease burden in affected individuals [1,4-6].

Despite their clinical relevance, attempts to attribute muscle symptoms to a single biological mechanism have been largely unsuccessful. Instead, symptom-based investigations have identified heterogeneous and partially overlapping symptom domains involving autonomic dysfunction, flu-like symptoms, cognitive impairment, sleep disturbances, and respiratory complaints. These findings suggest that muscle symptoms may arise from multiple interacting physiological processes rather than from isolated skeletal muscle pathology alone [7,8].

Several mechanistic domains have been implicated in the pathophysiology of ME/CFS, including persistent or intermittent immune activation, autonomic nervous system dysregulation, endothelial dysfunction, and abnormalities in cellular energy metabolism [9]. Impaired mitochondrial function, altered oxidative phosphorylation, reduced oxygen extraction, and abnormalities in neuromuscular performance have been reported [3], although findings have varied across patient cohorts and study methodologies. More recent evidence highlights disturbances in microcirculatory substrate delivery, nitric oxide-dependent vascular regulation, endothelial function, and autoantibodies targeting G-protein-coupled receptors (GPCRs), particularly β2-adrenergic and muscarinic receptors [4]. Together, these observations support the concept of a multisystem disorder in which impaired vascular and metabolic regulation may contribute substantially to muscle dysfunction and exercise intolerance [2,6,9].

Hormonal modulation adds a further layer of complexity. Estrogen influences vascular tone, endothelial function, immune signaling, and energy metabolism, and growing evidence suggests that symptom expression and disease trajectories differ between men and women. In this context, endocrine transitions such as menopause may alter symptom profiles and potentially influence the interaction between autonomic, immune, and vascular mechanisms in ME/CFS [10].

The present study builds on a series of analyses conducted within the APAV ME/CFS research program. In the first phase (Studies 1–2), we hypothesized that distinct symptom groups would emerge that correspond to established pathophysiological domains, particularly neuroimmune, autonomic, gastrointestinal, and central nervous system mechanisms. These analyses identified several biologically plausible symptom clusters that may facilitate improved patient stratification, support more precise diagnostic approaches, and inform the development of targeted therapeutic strategies [11,12].

In the subsequent phase (Studies 3–4), we investigated potential modifiers of symptom architecture, focusing on biological sex and menopausal status. Study 3 examined sex-related differences in symptom clusters associated with autonomic and central nervous system dysfunction, whereas Study 4 focused on gastrointestinal and immune-related symptom domains. Together, these studies demonstrated substantial sex-specific variation in symptom organization and provided further support for an integrated pathophysiological model of ME/CFS [13,14]. Now, the originally planned five-part study design was modified after the completion of Part 4. The results obtained thus far necessitated the expansion of the research program by two additional studies to ensure a comprehensive and systematic coverage of all relevant aspects. Accordingly, the final synthesis of the findings will be presented in Part 7 rather than in the originally envisaged Part 5 (see Figure 1).

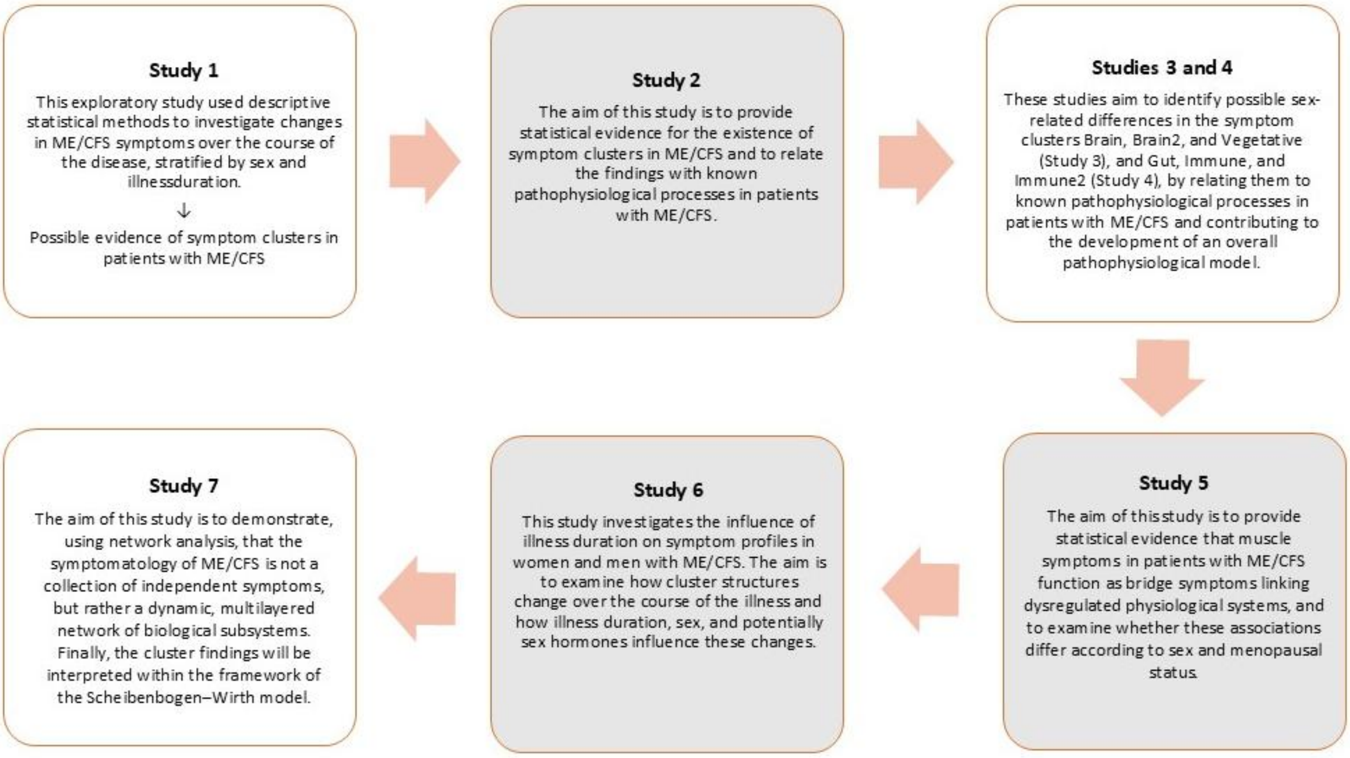

Figure 1 Chronological placement of Study 5 within the sequence of studies.

Against this background, the present study addresses a remaining question that emerged from the previous analyses: the role of muscle symptoms within the overall symptom architecture of ME/CFS. Although muscle complaints represent some of the most disabling manifestations of the disease, they did not consistently emerge as an independent symptom domain in earlier clustering analyses. This observation raises the possibility that muscle symptoms function as physiological bridge symptoms linking multiple dysregulated systems—including autonomic, vascular, immune, and metabolic pathways—rather than constituting a distinct pathophysiological entity. Such an interpretation would be consistent with contemporary models emphasizing systemic dysregulation and impaired physiological integration in ME/CFS [1,4].

The present study therefore complements the previous phases of the research program by specifically examining the position of muscle symptoms within the established symptom network and by exploring sex- and menopause-related differences in their manifestation.

The three objectives of this study (Study 5) are therefore:

1. To identify a physiological muscle symptom cluster within the APAV-ME/CFS dataset and determine whether muscle symptoms function as bridge symptoms across symptom domains.
2. To analyze sex- and menopause-specific differences in the structure and severity of muscle-related symptoms.
3. To integrate these findings into a mechanistic model linking symptom architecture with underlying autonomic, vascular, immunological, and metabolic pathways.

Methods

Study design and data source

This study represents the fifth phase of a series of symptom-architecture analyses conducted within the APAV-ME/CFS research program. The objective of the present analysis was to investigate the role of muscle-related symptoms within the broader symptom network of ME/CFS and to determine whether muscle symptoms form part of a distinct physiological dysregulation cluster. Particular emphasis was placed on sex- and menopause-related differences in symptom architecture.

The analyses were based on cross-sectional questionnaire data obtained from participants with physician-diagnosed ME/CFS who fulfilled the inclusion criteria of the APAV-ME/CFS registry. Details of recruitment procedures and the registry have been described previously (Habermann-Horstmeier and Horstmeier 2025, 2026).

- Symptom variables

Participants reported the presence or absence of a broad range of ME/CFS-related symptoms. For the present study, muscle problems served as the primary variable of interest.

The following symptom domains were initially examined as potential correlates of muscle problems:

- Brain fog
- Headaches
- Visual disturbances
- Flu-like symptoms
- Susceptibility to infection
- Gastrointestinal complaints
- Urogenital complaints
- (Food) Intolerances
- Cardiovascular symptoms
- Breathing problems
- Sleep disturbances
- Sensory hypersensitivities
- Temperature-regulation disturbances
- Temperature intolerance

All variables were coded dichotomously (present/absent).

- Sex and menopausal status

Sex was coded as female or male. The three individuals who identified as non-binary were excluded from the analysis prior to calculations due to the small group size. To investigate possible hormonal influences on symptom architecture, women were additionally stratified according to menopausal status. Stratification by menopausal status was, of course, restricted to female participants (n = 608) and was operationalized using an age-based cutoff (<50 vs. ≥50 years), which provides only a proxy for biological menopausal transition and does not capture perimenopausal fluctuations, hormonal variability, or the potential influence of hormone replacement therapy. Because detailed menopausal information was not collected, classification into pre- and postmenopausal groups remained approximate. Male participants (n = 137) were analyzed separately under the sex-stratified analyses but were not included in menopause-specific models.

- Disease duration

To explore the stability of the physiological symptom cluster over the course of illness, participants were stratified into three disease-duration categories:

- ≤ 2 years
- 3–8 years
- > 8 years

These categories were chosen to capture the early phase after disease onset, an intermediate phase, and long-standing chronic disease, allowing assessment of whether the latent physiological symptom architecture remained stable over the disease course.

Statistical analyses

All analyses were conducted using Stata 12.0.

- Logistic regression

The primary analytic approach consisted of multivariable logistic regression models with muscle problems as the dependent variable. An initial full model included all symptom domains listed above. Variables showing the strongest and most consistent associations with muscle problems were subsequently entered into a reduced physiological model.

Separate logistic regression analyses were performed for:

- the total sample,
- women and men separately,
- women classified as premenopausal and women classified as postmenopausal,
- disease-duration subgroups.

Statistical significance was defined as $p < 0.05$.

- Correlation analyses

Because all symptom variables were dichotomous, both Pearson correlations and tetrachoric correlations were calculated. Pearson correlations were used to provide descriptive estimates of symptom co-occurrence. Tetrachoric correlations were used to estimate latent associations between symptom domains under the assumption that dichotomous symptom variables represent underlying continuous traits.

- Exploratory factor analysis

Exploratory factor analysis was performed using the principal-factor method. Factor retention was guided by eigenvalues, factor interpretability, and inspection of loading patterns. Both unrotated and varimax-rotated solutions were examined. Factor loadings were interpreted as indicators of the extent to which individual symptoms contributed to shared latent symptom dimensions.

- Structural equation modeling

Structural equation modeling (SEM) was used to test whether the symptoms identified in the regression and factor analyses could be represented by a common latent physiological dysregulation factor. Muscle problems were specified as the reference indicator of the latent factor, and symptom domains showing consistent associations with muscle problems were entered as observed indicators. Model fit was evaluated using standard fit indices:

- Root Mean Square Error of Approximation (RMSEA)
- Comparative Fit Index (CFI)
- Tucker–Lewis Index (TLI)
- Standardized Root Mean Square Residual (SRMR)

Model fit was considered acceptable according to conventional criteria (RMSEA < 0.08, CFI > 0.90, TLI > 0.90, SRMR < 0.08).

- Analysis of gastrointestinal complaints

Because gastrointestinal complaints did not emerge as independent predictors in the primary regression models despite showing substantial correlations with muscle problems, additional analyses were performed to determine whether gastrointestinal complaints belonged to the same latent physiological structure. These analyses included:

- tetrachoric correlations,
- logistic regression models incorporating gastrointestinal symptoms,
- exploratory factor analysis,
- structural equation modeling.

The same analytical strategy was subsequently extended to exploratory analyses of urogenital symptoms within neurocognitive-autonomic symptom domains.

- Ethical considerations

The study analyzed anonymized questionnaire data collected within the APAV-ME/CFS research program. All procedures complied with applicable ethical and data-protection regulations. Participant confidentiality was maintained throughout the study.

- Stratified Training Dataset

To assess robustness, all analyses were repeated in a stratified, randomized training dataset derived from the full sample. Stratification was based on sex and illness duration to ensure proportional representation and minimize sampling bias. Results were directionally consistent across datasets.

Results

1. *Physiological correlates of muscle problems*

To identify symptom domains associated with muscle problems, a multivariable logistic regression including 14 symptom categories was performed. Among all variables entered into the model, breathing problems, flu-like symptoms, and temperature-regulation disorders remained significant independent predictors of muscle problems. Cardiovascular problems showed a positive but non-significant association. Neurocognitive symptoms (brain fog, headaches, sensory hypersensitivity), sleep disorders, susceptibility to infection, gastrointestinal complaints, urogenital symptoms, and (food) intolerance-related symptoms were not independently associated with muscle problems (Table 1).

Table 1. Multivariable logistic regression predicting muscle problems (full model, n = 745)

Predictor	**β**	**p-value**
Breathing problems	0.962	0.001
Flu-like symptoms	0.650	0.020
Temperature-regulation disorders	0.633	0.030
Sensory hypersensitivity	0.540	0.084
Sleep disorders	0.509	0.111
Cardiovascular problems	0.440	0.140
Visual disturbances	0.400	0.154
Brain fog	0.347	0.337
Temperature intolerance	0.266	0.362
Susceptibility to infection	0.217	0.436
Gastrointestinal complaints	0.066	0.827

Predictor	β	p-value
(Food) Intolerances	–0.148	0.616
Headaches	–0.219	0.427
Urogenital symptoms	–0.254	0.385

Model fit:

- LR $\chi^2 = 122.37$
- $p < 0.001$
- Pseudo-$R^2 = 0.211$

Abbreviations: n = number, β = Beta, regression coefficient; p-value = probability; LR χ^2 = Likelihood-Ratio-Chi-Quadrat-Test; R^2 = statistical measure of how closely the data fit the fitted regression line

To focus on the strongest symptom relationships, a reduced logistic model including breathing problems, flu-like symptoms, temperature-regulation disorders, cardiovascular problems, and visual disturbances was estimated. All five variables remained significant predictors of muscle problems. The strongest association was observed for breathing problems, followed by flu-like symptoms, temperature-regulation disorders, cardiovascular problems, and visual disturbances (Table 2).

Table 2. Reduced physiological model predicting muscle problems (n = 745; total sample)

Predictor	β	p-value
Breathing problems	1.121	<0.001
Flu-like symptoms	0.886	<0.001
Temperature-regulation disorders	0.822	0.002
Cardiovascular problems	0.654	0.017
Visual disturbances	0.516	0.046

Model fit:

- LR $\chi^2 = 111.37$
- $p < 0.001$
- Pseudo-$R^2 = 0.192$

Abbreviations: n = number; β = Beta, regression coefficient; p-value = probability; LR χ^2 = Likelihood-Ratio-Chi-Quadrat-Test; R2 = statistical measure of how closely the data fit the fitted regression line

2. *Correlation structure of the physiological symptom cluster*

Pearson correlations between muscle problems and the five core physiological symptoms ranged from 0.21 to 0.30. Tetrachoric correlations were substantially stronger, indicating robust associations at the latent level. Muscle problems correlated most strongly with breathing problems, cardiovascular problems, and temperature-regulation disorders, followed by visual disturbances and flu-like symptoms (Table 3).

Table 3. Tetrachoric correlations with muscle problems (total sample)

Symptom	Tetrachoric correlation
Breathing problems	0.539
Cardiovascular problems	0.501
Temperature-regulation disorders	0.495
Visual disturbances	0.408
Flu-like symptoms	0.394

3. *Factor analysis*

Factor analysis of the six physiological symptoms identified a dominant general factor. In the unrotated solution, breathing problems, cardiovascular problems, temperature-regulation disorders, muscle problems, visual disturbances, and flu-like symptoms all loaded positively on the first factor (Table 4). Following varimax rotation, breathing problems retained the strongest loading on Factor 1 (0.58), whereas flu-like symptoms showed a relatively stronger loading on Factor 2 (0.31), suggesting a secondary infection-like or PEM-related dimension.

4. *Structural equation modeling*

Structural equation modeling supported the existence of a latent physiological dysregulation factor underlying the symptom cluster. All six symptoms loaded significantly on the latent factor. Breathing problems showed the strongest loading, followed by visual disturbances, cardiovascular problems, temperature-regulation disorders, and flu-like symptoms. Model fit was good, indicating that a single latent physiological factor adequately accounted for the observed symptom relationships (Table 4).

Table 4. Factor analysis and SEM loadings of the physiological dysregulation factor (total sample)

Symptom	Factor loading	SEM loading
Breathing problems	0.587	1.764
Cardiovascular problems	0.552	1.344
Temperature-regulation disorders	0.505	1.249
Muscle problems	0.490	1.000 (fixed)
Visual disturbances	0.467	1.382
Flu-like symptoms	0.298	0.725

SEM fit:

- RMSEA = 0.046
- CFI = 0.972
- TLI = 0.953
- SRMR = 0.026

Abbreviations: RMSEA = Root Mean Square Error of Approximation; CFI = Comparative Fit Index; TLI = Tucker-Lewis Index; SRMR = Standardized Root Mean Square Residual; CD = Coefficient of Determination; SEM = Structural equation modeling

5. Sex-specific analyses

Among women, all five physiological symptoms remained significant predictors of muscle problems. Flu-like symptoms, breathing problems, cardiovascular problems, temperature-regulation disorders, and visual disturbances contributed independently to the model. In men, breathing problems emerged as the dominant predictor of muscle problems, with temperature-regulation disorders also remaining significant. Flu-like symptoms, cardiovascular problems, and visual disturbances were not independently associated with muscle problems (Table 5 A, B).

6. Menopause-specific analyses

Distinct symptom patterns were observed according to menopausal status. Among women classified as premenopausal (n = 349), cardiovascular problems, visual disturbances, and temperature-regulation disorders independently predicted muscle problems, whereas flu-like symptoms and breathing problems did not reach significance. In women classified as postmenopausal (n = 259), the pattern shifted markedly: breathing problems and flu-like symptoms emerged as the only significant predictors of muscle problems, while cardiovascular problems, temperature-regulation disorders, and visual disturbances lost their independent predictive value (Table 5 C, D).

Despite these differences in regression models, tetrachoric correlations, factor analyses, and SEM analyses demonstrated a largely comparable latent physiological factor structure in both menopausal groups. Tetrachoric correlations between muscle problems and the core physiological symptoms were of similar magnitude in both groups, with the strongest correlation observed for cardiovascular problems in women classified as premenopausal ($\rho = 0.61$) and for breathing problems in women classified as postmenopausal ($\rho = 0.55$). SEM analyses confirmed that all six physiological symptoms loaded significantly on a single latent physiological dysregulation factor in both groups. Model fit was excellent in the premenopausal subgroup ($\chi^2(9) = 14.51$, $p = 0.105$; RMSEA = 0.042, 90 % CI: 0.000–0.080; CFI = 0.976; TLI = 0.961; SRMR = 0.028) but only borderline acceptable in the postmenopausal subgroup ($\chi^2(9) = 25.87$, $p = 0.002$; RMSEA = 0.085, 90 % CI: 0.048–0.124; CFI = 0.901; TLI = 0.835; SRMR = 0.048). The reduced fit in the postmenopausal group—particularly the TLI below 0.90 and the RMSEA slightly above the conventional 0.08 threshold—suggests that the strictly unidimensional model captures the symptom structure less precisely in this subgroup, possibly because of emerging secondary substructure or smaller sample size (Table 5 E).

Breathing problems consistently showed the strongest SEM loadings in women classified as postmenopausal (loading = 1.87) compared to women classified as premenopausal (loading = 1.49). In contrast, flu-like symptoms exhibited somewhat stronger loadings on the latent factor in women classified as premenopausal (SEM loading = 0.73) than in women classified as postmenopausal (SEM loading = 0.60), and a corresponding pattern was observed in the unrotated factor analyses (premenopausal Factor-1 loading = 0.32; postmenopausal Factor-1 loading = 0.23). Cardiovascular problems showed a stronger SEM loading in women classified as premenopausal (1.38 vs. 1.11), whereas visual disturbances loaded more strongly in women classified as postmenopausal (1.50 vs. 1.35); temperature-regulation disorders showed comparable loadings in both groups (1.25 vs. 1.32) (Table 5 F).

Notably, this divergence between factor loadings and regression coefficients for flu-like symptoms reflects a substantively meaningful pattern. In women classified as premenopausal, flu-like symptoms loaded substantially on the shared physiological factor but did not contribute an independent regression signal—presumably because their variance was largely shared with cardiovascular, visual, and thermoregulatory symptoms in this age group. In women classified as postmenopausal, by contrast, the independent contributions of cardiovascular, visual, and thermoregulatory symptoms decreased, allowing flu-like symptoms to emerge as a partially independent predictor of muscle problems despite a slightly lower factor loading. This pattern indicates that menopausal status not only shifts the relative prominence of individual symptom domains within the same underlying physiological dysregulation factor but also alters how these domains share or partition variance with respect to muscle symptoms.

Table 5. Sex- and menopause-specific predictors of muscle problems and SEM-Fit Overview (women classified as premenopausal and women classified as postmenopausal); A and B: total sample; C, D, E and F: sample women only.

A. Women (n = 608)

Predictor	**β**	**p**
Flu-like symptoms	0.967	0.001
Breathing problems	0.850	0.004
Cardiovascular problems	0.783	0.012
Temperature-regulation disorders	0.759	0.013
Visual disturbances	0.737	0.011

B. Men (n = 137)

Predictor	**β**	**p**
Breathing problems	3.489	0.002
Temperature-regulation disorders	1.426	0.017
Flu-like symptoms	0.716	0.242
Cardiovascular problems	0.037	0.953
Visual disturbances	–0.571	0.353

C. Women classified as premenopausal (n = 349)

Predictor	**β**	**p**
Cardiovascular problems	1.333	0.002
Visual disturbances	1.192	0.004
Temperature-regulation disorders	0.888	0.035
Flu-like symptoms	0.431	0.322
Breathing problems	0.424	0.311

D. Women classified as postmenopausal (n = 259)

Predictor	**β**	**p**
Breathing problems	1.328	0.003
Flu-like symptoms	1.194	0.002
Cardiovascular problems	0.323	0.515
Temperature-regulation disorders	0.604	0.177
Visual disturbances	0.359	0.418

E. SEM-Fit Overview (women classified as premenopausal and women classified as postmenopausal)

Subgroup	n	$\chi^2(9)$	p	RMSEA (90% CI)	CFI	TLI	SRMR	CD
Premenopausal	349	14.51	0.105	0.042 (0.000–0.080)	0.976	0.961	0.028	0.687
Postmenopausal	259	25.87	0.002	0.085 (0.048–0.124)	0.901	0.835	0.048	0.683

F. SEM-Loadings (women classified as premenopausal and women classified as postmenopausal)

Symptom	**Premenopausal (n = 349)**	**Postmenopausal (n = 259)**
Muscle problems	1.00 (fixed)	1.00 (fixed)
Flu-like symptoms	0.73	0.60
Breathing problems	1.49	1.87
Cardiovascular problems	1.38	1.11
Temperature-regulation disorders	1.25	1.32
Visual disturbances	1.35	1.50

Abbreviations: n = number; β = Beta, regression coefficient; p-value = probability; RMSEA = Root Mean Square Error of Approximation; CFI = Comparative Fit Index; TLI = Tucker-Lewis Index; SRMR = Standardized Root Mean Square Residual; CD = Coefficient of Determination

7. *Influence of disease duration*

The physiological symptom cluster remained stable across disease-duration groups. Breathing problems, temperature-regulation disorders, cardiovascular problems, visual disturbances, and flu-like symptoms continued to show positive associations with muscle problems across all duration categories. SEM analyses demonstrated acceptable to excellent model fit in all disease-duration groups, indicating persistence of the latent physiological dysregulation factor throughout the course of illness.

8. Gastrointestinal and urogenital symptoms

In the full model, gastrointestinal complaints showed a significant tetrachoric correlation with muscle problems ($\rho = 0.39$, $p < 0.001$). Although gastrointestinal complaints did not independently predict muscle problems in multivariable regression ($\beta = 0.20$, $p = 0.469$), they loaded substantially on the physiological factor in factor analysis (loading = 0.52) and showed a strong loading in SEM (SEM loading = 1.48, $p < 0.001$). Inclusion of gastrointestinal complaints produced a well-fitting latent-factor model (RMSEA = 0.047, CFI = 0.966, SRMR = 0.028) (Table 6).

In contrast, urogenital symptoms showed no significant association with muscle problems in regression analyses and were not included in the final physiological-factor models.

Table 6. Gastrointestinal complaints and the physiological dysregulation factor (total sample)

Analysis	**Result**
Tetrachoric correlation with muscle problems	0.392
Logistic regression	$\beta = 0.196$, $p = 0.469$
Factor loading	0.517
SEM loading	1.478
RMSEA	0.047
CFI	0.966

Abbreviations: β = Beta; regression coefficient; p-value = probability; SEM = Structural Equation Modeling; RMSEA = Root Mean Square Error of Approximation; CFI = Comparative Fit Index

Discussion

1. Core findings

The present study identifies a robust and coherent physiological symptom cluster in ME/CFS, with muscle problems at its core. Across multiple analytic approaches—including logistic regression, tetrachoric correlations, factor analysis, and structural equation modeling—muscle problems consistently co-occurred with breathing difficulties, cardiovascular symptoms, temperature dysregulation, visual disturbances, flu-like symptoms and, in secondary analyses, with gastrointestinal complaints. Notably, gastrointestinal complaints showed a level of latent association with muscle symptoms comparable to that of flu-like symptoms and visual disturbances and loaded strongly on the same physiological factor despite lacking independent predictive value in regression analyses. The convergence of findings across methods provides strong evidence that these symptoms reflect a shared underlying mechanism rather than independent or coincidental co-occurrences [9,12,15].

2. Integration with previous ME/CFS literature

Mitochondrial studies reported alterations in mitochondrial membrane potential, reduced ATP production rates, and impaired oxidative phosphorylation in subsets of patients, although findings have been heterogeneous across cohorts [6,7]. In parallel, metabolomic analyses have revealed altered acylcarnitine profiles and disruptions in acyl-lipid metabolism, consistent with impaired pyruvate dehydrogenase function and inefficiencies in aerobic energy metabolism and substrate utilization [16-18], with emerging evidence that the magnitude and pattern of these metabolic alterations may differ between male and female patients [10].

Beyond intrinsic metabolic alterations, several studies have highlighted abnormalities in neuromuscular activation. In particular, reduced voluntary activation and altered perception of effort have been reported, suggesting that central neural mechanisms may contribute to exertional intolerance and muscle fatigability—although the available evidence base remains limited [3]. Autonomic and vascular dysfunction—including endothelial dysfunction, reduced nitric-oxide bioavailability, impaired vasodilation alongside excessive sympathetic vasoconstriction, and reduced tissue perfusion—has also been implicated, linking muscle symptoms to broader disturbances in vascular and autonomic regulation [4, 19-23]—pathways that are themselves modulated by sex-hormone signaling and by menopause-related vascular and autonomic changes [24-26].

Together, these findings established a framework in which muscle symptoms in ME/CFS arise from a combination of impaired energy metabolism, altered neuromuscular activation, and dysregulated perfusion [1,2,9]. The recent mechanistic studies presented at the 2026 International ME/CFS Conference in Berlin refine this framework by identifying specific metabolic, endothelial, and immunological pathways—such as alterations in skeletal-muscle substrate metabolism, arginine–NO and polyamine dysregulation, GPCR-targeting autoantibodies, and arachidonic-acid pathway disturbances [27-30]—and by revealing sex- and menopause-dependent differences in the clinical expression of these processes. Such sex- and hormone-related modulations of immune, vascular, and autonomic regulation [10,31] are increasingly recognized as a relevant dimension of ME/CFS pathophysiology and are examined in detail in later sections of this discussion [14].

3. *Bridge symptoms*

Several symptoms in our dataset appear to function as bridge symptoms, linking multiple symptom domains and reflecting the downstream expression of interacting physiological processes rather than isolated organ-specific dysfunction. Muscle problems represent the clearest example. Although they load most strongly on the muscle factor, they also show substantial associations with autonomic, breathing-related, cardiovascular, and flu-like symptoms. This pattern suggests that muscle symptoms integrate neuro-metabolic, vascular, autonomic, and immunological mechanisms. Such an interpretation is compatible with recent evidence for impaired microcirculatory substrate delivery [27], altered oxygen extraction [23], reduced NO-dependent vasodilation [28], GPCR-targeting autoantibodies affecting autonomic signalling [29,32], and dysregulated arachidonic-acid metabolism [30]. Together with known abnormalities in neuromuscular activation and autonomic regulation of muscle perfusion [1,22], these findings position muscle symptoms as a prototypical neuro-metabolic-vascular bridge symptom in ME/CFS.

Additional bridge symptoms may include shortness of breath, temperature dysregulation, cognitive slowing, visual disturbances, and generalized fatigue. These symptoms span multiple physiological domains and are compatible with shared mechanisms involving autonomic regulation [26], microcirculatory function, inflammatory signalling, and metabolic stress responses. This interconnected symptom architecture supports the interpretation that the major symptom clusters observed in ME/CFS arise from partially overlapping biological processes rather than from isolated dysfunction within individual organ systems.

4. *Alternative Interpretation: The Physiological Dysregulation Factor as a Manifestation of a Chronic PEM-Associated Danger Response*

An alternative interpretation, which is compatible with our findings, is to view the identified physiological dysregulation factor through the lens of post-exertional malaise (PEM). PEM is widely regarded as the hallmark symptom of ME/CFS and can be triggered by physical, cognitive, orthostatic, or emotional exertion [1,4-6]. Regardless of the nature of the trigger, many patients describe a state resembling a systemic illness response characterized by profound fatigue, muscle pain, muscle weakness, chills, flu-like symptoms, and a worsening of multiple symptom domains. This symptom constellation closely resembles the evolutionarily conserved "sickness response" [33], a coordinated physiological program normally activated in response to infection, tissue injury, or other forms of biological threat.

From this perspective, PEM may represent a maladaptive or chronically activated form of such a danger or illness response. Whereas an acute sickness response is typically self-limiting and resolves once the triggering stimulus has been eliminated, ME/CFS may involve a state in which the underlying signaling pathways fail to fully deactivate or become excessively responsive to repeated stressors. The result could be a persistent tendency to activate a physiological program that is normally protective but becomes functionally disabling when chronically engaged.

Within such a framework, muscle-related symptoms occupy a particularly important position. Muscle tissue is among the most metabolically active tissues in the body and is highly sensitive to alterations in energy availability, oxygen delivery, inflammatory signaling, and systemic stress responses. Commonly reported symptoms such as myalgia, weakness, heaviness, and delayed recovery following exertion can therefore be understood as prominent manifestations of a systemic illness response. Importantly, this response would not be restricted to the skeletal muscles of the limbs. Rather, it could affect all muscle systems throughout the body, including respiratory muscles, cardiac muscle, vascular smooth muscle, and other muscle-dependent physiological functions. Such a model provides a biologically coherent explanation for why muscle-related symptoms in our analyses were closely associated with respiratory complaints, cardiovascular symptoms, temperature regulation disturbances, and other physiological symptom domains. Interestingly, gastrointestinal symptoms also appeared to belong to this broader physiological network. Although they did not independently predict muscle problems in regression analyses, they showed substantial correlations with muscle symptoms and loaded strongly on the latent physiological dysregulation factor. This pattern suggests that gastrointestinal manifestations may represent another downstream expression of the same systemic process rather than an independent symptom domain. Such an interpretation is compatible with the concept of PEM as a body-wide illness response affecting multiple organ systems simultaneously.

From this viewpoint, muscle symptoms may represent the most visible clinical expression of a broader systemic response. Their role as bridge symptoms would arise from their position at the intersection of multiple physiological systems that react to the same underlying danger and stress signals. The latent physiological dysregulation factor identified in our analyses may therefore reflect the symptomatic manifestation of a chronic PEM-associated illness response involving several organ systems simultaneously.

Importantly, this framework does not replace autonomic, vascular, immunological, or metabolic mechanisms. Rather, it provides an overarching conceptual model within which these processes may interact. The particularly strong associations observed in our data between muscle symptoms, respiratory complaints, cardiovascular manifestations, and temperature regulation disturbances continue to support a substantial contribution of autonomic and vascular regulatory pathways. The present interpretation therefore views these mechanisms not as competing explanations but as potential biological mediators of a maladaptive systemic danger response. In this sense, the various symptom clusters observed in ME/CFS may represent different manifestations of a common stress-responsive pathophysiological network rather than entirely independent disease processes.

5. *Sex- and menopause-specific symptom architecture*

One of the key contributions of this work is the demonstration that the structure and drivers of the physiological muscle cluster differ between women classified as premenopausal, women classified as postmenopausal, and men. Both female subgroups exhibited physiological dysregulation patterns, but the dominant symptom associations differed. Women classified as premenopausal showed stronger links with cardiovascular symptoms, visual disturbances, and temperature-regulation problems, whereas women classified as postmenopausal displayed a more compact symptom pattern characterized primarily by breathing difficulties and flu-like symptoms. Men display a symptom pattern dominated by breathing problems and temperature-regulation symptoms. The postmenopausal symptom architecture shows partial convergence toward the male phenotype [31,34,35], particularly with regard to the increasing prominence of breathing-related symptoms. However, unlike men, women classified as postmenopausal continued to exhibit a significant association between muscle problems and flu-like symptoms, suggesting persistence of either an immunological component or a symptom complex related to post-exertional malaise (PEM).

The observed differences between the premenopausal-group and the postmenopausal-group suggest that hormonal status may influence the manifestation of physiological dysregulation in ME/CFS. Estrogen is known to affect immune responsiveness, autonomic regulation, vascular function, and thermoregulation [24-26]. Consequently, menopause may alter the relative contribution of these mechanisms to symptom expression, even when the overall physiological dysregulation factor remains present. To be precise, this means that hormonal changes shift the relative importance of individual expressions of this factor. An additional finding of this study is that women classified as postmenopausal show several features [31,34,35] that resemble the male phenotype [21-23,27]. Both groups demonstrated strong associations between muscle problems and breathing-related symptoms, suggesting a greater prominence of respiratory and possibly autonomic-perfusion-related manifestations within the shared physiological dysregulation factor. However, the continued association between muscle problems and flu-like symptoms in women classified as postmenopausal suggests that convergence toward the male phenotype is only partial.

Recent experimental evidence from passive-transfer models further supports the existence of sex-specific disease mechanisms in ME/CFS and related post-infectious syndromes. Patient-derived IgG has been shown to induce sensory and autonomic dysfunction in experimental systems, suggesting that pathogenic autoantibodies may contribute directly to neuroimmune dysregulation [36]. Emerging evidence further indicates that autoantibody repertoires and their neurobiological effects may differ between women and men and may vary across the female lifespan [37]. Together, these findings suggest that hormonal status may influence not only symptom expression but also the underlying biological pathways contributing to disease manifestations.

An alternative explanation for the persistence of flu-like symptoms in women classified as postmenopausal is that this symptom category may partially capture the subjective experience of PEM rather than classical infection-like immune activation. Notably, susceptibility to infection—the second component of the immune symptom cluster identified in our previous analyses—was not associated with muscle symptoms in any of the regression models. This selective association of flu-like symptoms, but not infection susceptibility, argues against a simple interpretation in terms of generalized immune vulnerability and is compatible with the possibility that flu-like symptoms partly capture the experiential and physiological manifestations of post-exertional malaise. Additionally, many patients describe PEM as a state characterized by profound chills, internal coldness, shivering sensations, and generalized illness resembling influenza despite the absence of overt infection [38]. Beyond their flu-like quality, PEM episodes are frequently described in terms that point directly to muscle involvement. The IOM report [39] cites patient accounts in which individuals describe their arms, legs, and entire body as feeling heavy and harder to move, and compare themselves to a battery that can never be fully recharged despite extensive rest and a reduction of activities to the bare essentials. Such descriptions illustrate that the subjective experience commonly labelled 'fatigue' in ME/CFS is often phenomenologically a muscle experience—one of heaviness, weakness, and an inability to mobilize force or recover after minimal exertion. From this perspective, muscle symptoms are not merely one component of PEM among others but a central sensory and functional dimension of how PEM is experienced. This phenomenological closeness between fatigue, muscle symptoms, and flu-like sensations adds an experiential dimension to the bridge-symptom interpretation discussed above and supports the view that flu-like symptoms in women classified as postmenopausal partly reflect PEM-associated muscle and autonomic phenomena rather than a discrete immune-activation profile—in line with the broader interpretation of PEM as a chronic, systemically expressed illness response outlined above.

These interpretations are not mutually exclusive. Autonomic, vascular, immunological, and metabolic abnormalities may represent biological mediators through which a chronic PEM-associated illness response becomes clinically expressed.

6. *Hormonal and biological interpretation*

The shift in symptom architecture across menopausal groups—from a predominantly cardiovascular-visual-thermoregulatory pattern in women classified as premenopausal toward a breathing- and flu-like-dominated pattern in women classified as postmenopausal —may be driven by hormone-dependent changes in vascular and autonomic regulation. Estrogen is a potent modulator of endothelial nitric oxide (NO) production, microvascular tone, and capillary density. Its decline after menopause reduces endothelial NO bioavailability, impairs vasodilation [40], and increases vascular stiffness [24,25], thereby limiting skeletal muscle perfusion and substrate delivery. These changes are compatible with the reduced intramuscular glucose availability observed in microdialysis studies [27] and with the strong association between muscle problems and breathing- as well as flu-like symptoms in our postmenopausal subgroup.

The menopause transition may also represent a particularly important window for neuroimmune dysregulation. During the pre- and perimenopausal years, fluctuating estradiol concentrations can enhance B-cell activation, antibody production, and inflammatory responsiveness, whereas the concomitant decline in progesterone may reduce neuroprotective and immunoregulatory buffering mechanisms. Such hormonal dynamics [41] could facilitate the generation and persistence of pathogenic autoantibodies and amplify neuroimmune signalling pathways [42]. Genetic factors may further contribute to this susceptibility. In women, incomplete X-chromosome inactivation can result in increased expression of immune-related genes [43], including TLR7 [44], thereby promoting heightened immune responsiveness and a greater propensity for autoantibody formation [45,46]. While our study does not directly assess these mechanisms, they provide a biologically plausible background for the sex- and menopause-specific symptom configurations observed in our data.

Within this framework, hormonal changes associated with menopause may alter the relative prominence of specific symptom domains without abolishing the underlying physiological dysregulation factor. Estrogen withdrawal may enhance sympathetic nervous system activity, reduce endothelial adaptability, and impair microvascular regulation. Such changes are consistent with the increased prominence of breathing-related symptoms observed in women classified as postmenopausal and may partly explain their partial convergence toward the symptom profile seen in men—specifically with regard to the dominance of breathing problems. However, unlike men, women classified as postmenopausal retained a substantial contribution of flu-like symptoms. Indeed, flu-like symptoms emerged as an independent regression predictor of muscle problems only in this subgroup ($\beta = 1.19$, $p = 0.002$), whereas in women classified as premenopausal they loaded substantially on the latent physiological factor but did not reach independent significance—presumably because their variance was largely shared with cardiovascular, visual, and thermoregulatory symptoms in that age group. This nuanced pattern indicates that menopause does not simply produce a male-like phenotype but rather a distinct configuration in which immune-related and autonomic-vascular features coexist and partly redistribute their statistical signature.

A further line of evidence supports this interpretation at the level of model structure. The strictly unidimensional latent physiological dysregulation factor fit the data excellently in women classified as premenopausal (RMSEA = 0.042; CFI = 0.976; TLI = 0.961) but only borderline-acceptably in women classified as postmenopausal (RMSEA = 0.085; CFI = 0.901; TLI = 0.835) and in men (RMSEA = 0.097; CFI = 0.900; TLI = 0.833). The premenopausal group thus shows the cleanest statistical expression of a single underlying physiological dysregulation process, whereas women classified as postmenopausal and men display additional substructure that the unidimensional model captures less precisely.

The prominent emergence of flu-like symptoms as an independent predictor in women classified as postmenopausal may reflect continued or even enhanced involvement of immune-related pathways, altered stress-response mechanisms, or aspects of PEM that are experienced as a flu-like illness state (see Section 5). At the same time, the stronger independent contributions of cardiovascular, visual, and thermoregulatory symptoms observed in women classified as premenopausal suggest that hormonal status influences which manifestations of the shared physiological dysregulation factor become most clinically prominent at a given life stage—rather than determining whether the factor is present at all.

Collectively, these observations support the view that sex hormones modulate the expression of a common underlying physiological dysregulation process rather than driving a simple transition from an immune-dominant to an autonomic-vascular phenotype. This interpretation is further supported by the persistence of a comparable latent factor structure across menopausal groups and by the continued inclusion of gastrointestinal symptoms within the same physiological network despite substantial shifts in the relative prominence of individual symptom domains.

An interesting clinical observation that may be compatible with this framework concerns restless legs syndrome (RLS). Epidemiological studies consistently report a substantially higher prevalence of RLS in women than in men, although the biological basis of this sex difference remains incompletely understood [47,48]. It is conceivable that, in a subset of patients, RLS reflects aspects of the same autonomic, vascular, and metabolic dysregulation [49] proposed here as the underlying physiological dysregulation factor. Sex-specific differences in adipose tissue distribution [50], adrenergic receptor expression, and microvascular regulation may contribute both to female predominance of RLS and to its overlap with disorders characterized by impaired autonomic and vascular regulation [51]. Although the present data do not address this hypothesis directly, it may warrant investigation in future mechanistic studies.

7. *Mechanistic support from 2026 conference studies*

Recent microdialysis data presented at the 2026 International ME/CFS Conference in Berlin [27] provide preliminary physiological support for the symptom patterns observed in our postmenopausal and male subgroups. Despite normal systemic insulin sensitivity and normal systemic glucose and lipid responses, ME/CFS patients showed markedly reduced glucose availability in skeletal muscle during oral glucose challenge. This pattern suggests impaired microcirculatory substrate delivery rather than exclusively intrinsic metabolic dysfunction. The lower lactate/pyruvate ratio further suggests preserved or enhanced oxidative glucose utilization, arguing against a dominant defect in glycolytic pathways.

These findings align closely with our symptom-based analyses. In women classified as postmenopausal and men, muscle problems were most strongly associated with breathing difficulties and, depending on subgroup, with flu-like or temperature-regulation. Such a constellation may reflect the combined effects of impaired perfusion, autonomic dysregulation, and broader stress-response mechanisms, all of which are compatible with the physiological dysregulation factor identified in our study. The microdialysis data therefore provide a plausible biological substrate for this symptom architecture.

Independent support for biologically distinct ME/CFS subtypes comes from recent GPCR autoantibody profiling in a large post-infectious cohort presented by Schmitz et al. at the 2026 International ME/CFS Conference in Berlin. Hierarchical clustering of autoantibodies against vasoregulatory, autonomic, and neuro-immunological receptors identified four patient subgroups with distinct biological signatures. These clusters differed systematically in fatigue severity, muscle strength, orthostatic symptoms, endothelial dysfunction (RHI $\leq$ 1.8), and autonomic symptom burden. Autoantibodies targeting β-adrenergic, α-adrenergic, angiotensin II, endothelin, and muscarinic receptors have been associated with symptom severity and autonomic dysfunction in ME/CFS, with receptor-specific patterns differing between patient subgroups [52]. Earlier studies and reviews identified dysregulation of β-adrenergic and muscarinic receptor autoantibodies and proposed their involvement in autonomic and vascular dysfunction [4, 53], while Cabral-Marques et al. [54] demonstrated broader GPCR autoantibody signatures associated with immune homeostasis. Autoantibodies targeting β-adrenergic, α-adrenergic, angiotensin, endothelin, muscarinic, and nicotinic receptors showed differential associations with symptom severity, indicating subgroup-specific dysregulation of autonomic and vascular control mechanisms [4,53,54]. These findings align with our observation of distinct physiological dysregulation patterns and support the presence of mechanistically distinct ME/CFS subtypes.

Additional support for a shared physiological pathway in ME/CFS comes from recent multi-omics data presented by Dros et al. at the 2026 International ME/CFS Conference in Berlin [30]. In adolescents with post-acute infectious syndromes (ME/CFS, QFS, and Post-COVID), untargeted metabolomics revealed consistent reductions in LTB4-like arachidonic-acid metabolites, including several DiHETEs and leukotriene derivatives. These metabolites are important regulators of microvascular tone, endothelial function, and inflammatory signalling. Their downregulation across PAIS suggests a common disturbance in arachidonic-acid-dependent vascular and immunometabolic pathways. This observation is compatible with our finding that breathing-related and thermoregulatory manifestations become relatively more prominent in women classified as postmenopausal and men.

Collectively, these independent studies converge on a pathophysiological framework characterized by impaired microcirculation, autonomic dysregulation, altered endothelial adaptability, and disturbed immunometabolic signalling, broadly consistent with the physiological dysregulation factor identified in our study (Table 7).

Taken together, the emerging evidence is compatible with a broader framework of multisystemic neuro-immune dysregulation (MuNID), in which immune activation, autoantibody-mediated neuronal interference, autonomic dysfunction, vascular dysregulation, and impaired cellular energy metabolism interact as components of a dynamic pathophysiological network [15,55]). Within this model, sex and hormonal status may influence the clinical expression of this network rather than defining entirely distinct disease mechanisms. Consistent with our findings, women classified as premenopausal exhibited stronger associations with cardiovascular symptoms, visual disturbances, and temperature-regulation disorders, whereas men and women classified as postmenopausal showed greater prominence of breathing-related symptoms. The persistence of flu-like symptoms in women classified as postmenopausal further suggests that hormonal status may modify the relative visibility of specific symptom domains while preserving the underlying physiological dysregulation factor. The convergence of symptom-based analyses with recent immunological, metabolomic [17], and vascular studies supports the concept that ME/CFS is best understood as a heterogeneous but interconnected neuroimmune disorder rather than as an isolated dysfunction of a single physiological system.

Table 7 Summary of the core mechanisms identified in the studies presented at the 2026 International ME/CFS Conference in Berlin that pertain to muscle symptoms, and how they align with our model

Study	Biological Axis	Core Mechanism Identified	Alignment With Our Model
Jauert et al. (2026) [27]	Skeletal muscle microdialysis	Reduced intramuscular glucose availability despite normal systemic insulin sensitivity; preserved lipolysis; normal or lower lactate/pyruvate ratio → microcirculatory substrate and oxygen delivery deficit	Explains muscle symptoms, breathing problems, temperature dysregulation; supports the physiological dysregulation factor
Pipper-Krampl (2026) [28]	Arginine–NO and polyamine axis	Male-specific reductions in arginine, homo-arginine, citrulline and spermidine → reduced NO-dependent vasodilation, impaired endothelial compensation	Mechanistic basis for perfusion-related symptom manifestations observed in men and women classified as postmenopausal
Schmitz et al. (2026) [29]	GPCR-targeting autoantibodies	Four AAB-defined subgroups with distinct autonomic and vasoregulatory receptor profiles; correlations with fatigue, muscle strength, orthostatic symptoms, endothelial dysfunction	Independent evidence for autonomic-vascular subtypes and symptom-linked receptor dysregulation
Dros et al. (2026) [30]	Arachidonic-acid metabolism	Consistent reductions in LTB4-like metabolites and DiHETEs across PAIS; dysregulated lipoxygenase and CYP-450 pathways → impaired vascular and immunometabolic signaling	Supports a shared perfusion-related end-mechanism across post-infectious syndromes

8. *Gastrointestinal and urogenital symptoms*

Beyond the core physiological cluster, we examined whether gastrointestinal and urogenital symptoms should be considered part of the same latent structure. Gastrointestinal symptoms showed a moderate tetrachoric correlation with muscle problems and loaded substantially on the same general physiological factor in the factor analysis. In the SEM, gastrointestinal symptoms demonstrated a strong and highly significant loading on the latent physiological dysregulation factor, comparable in magnitude to cardiovascular and temperature-regulation symptoms. Notably, the tetrachoric correlation between gastrointestinal complaints and muscle problems ($\rho = 0.39$) was nearly identical to that observed for flu-like symptoms ($\rho = 0.39$), while the SEM loading of gastrointestinal symptoms (1.48) was among the strongest within the entire physiological factor model. Although gastrointestinal complaints did not independently predict muscle problems in multivariable regression—likely due to shared variance with breathing, cardiovascular, and thermoregulatory symptoms—their consistent loadings across analytic methods indicate that they represent a meaningful component of the broader physiological dysregulation pattern.

In contrast, urogenital symptoms showed only weak correlations with muscle problems, did not reach significance in regression models, and exhibited low factor loadings with high uniqueness values, indicating that they do not share the same underlying variance structure. Their SEM loading, although statistically significant due to model constraints, was accompanied by high residual variance, suggesting that urogenital complaints may arise from partially distinct mechanisms—such as pelvic floor dysfunction, local mucosal inflammation, or mast-cell activation—rather than from the systemic autonomic-metabolic processes that characterize the physiological muscle cluster.

An alternative explanation for the distinct behavior of urogenital symptoms is that they may be linked more closely to neuro-autonomic regulatory disturbances than to the physiological dysregulation factor centered on muscle symptoms. In additional analyses, urogenital complaints showed substantial associations with brain fog, sensory hypersensitivity, visual disturbances, sleep disorders, and temperature-regulation problems, and loaded significantly on a common latent neuro-autonomic factor. Such a pattern is compatible with disturbances of central and autonomic control mechanisms involved in lower urinary tract function, including sympathetic, parasympathetic, and somatic pathways. Although the present data do not permit conclusions regarding specific neural structures, they suggest that urogenital symptoms may reflect dysfunction of regulatory networks governing bladder and pelvic-organ function rather than the multisystem physiological processes underlying the muscle-centered dysregulation cluster.

Taken together, these findings are compatible with the inclusion of gastrointestinal symptoms, but not urogenital symptoms, in the conceptual model of physiological dysregulation in ME/CFS.

9. *Strengths and limitations*

The study has several strengths, including the use of multiple complementary statistical methods, the consistency of findings across analytic approaches, and the explicit stratification by menopausal status. The identification of a latent physiological dysregulation factor with excellent model fit in the total sample further strengthens the interpretation that these symptoms share a common underlying mechanism. Subgroup analyses by sex, menopausal status, and disease duration enabled a differentiated view of how this shared dysregulation factor is clinically expressed across patient groups.

Several limitations should be acknowledged. The cross-sectional design precludes causal inference, and the dichotomization of symptoms may obscure gradations of severity. Subgroup analyses, particularly in women classified as postmenopausal and men, were based on smaller sample sizes (n = 259 and n = 137, respectively) and should therefore be interpreted as exploratory. In addition, several mechanistic findings discussed here derive from conference presentations and await peer-reviewed publication and independent replication. Future work should examine whether these physiological clusters predict functional outcomes, disease trajectories, or treatment responses, and whether they can be replicated in independent cohorts.

A further limitation relates to the operationalization of menopausal status, which was based on an age-based proxy (< 50 vs. ≥ 50 years) rather than on direct hormonal measurements or detailed reproductive history. This approach does not capture perimenopausal fluctuations, individual variability in the timing of menopause, or the potential influence of hormone replacement therapy. Consequently, the observed differences between women classified as premenopausal and women classified as postmenopausal groups likely underestimate the true magnitude of hormone-related effects on symptom architecture. Future studies should employ more refined classifications—incorporating hormone levels, menstrual history, and information on hormone replacement therapy—to clarify the hormonal mediators underlying the observed patterns.

With regard to the structural equation models, fit was excellent for the total sample (RMSEA = 0.046; CFI = 0.972), for the female subgroup, and particularly for women classified as premenopausal (RMSEA = 0.042; CFI = 0.976). However, fit was only borderline acceptable in women classified as postmenopausal (RMSEA = 0.085; CFI = 0.901; TLI = 0.835) and in men (RMSEA = 0.097; CFI = 0.900; TLI = 0.833). The reduced fit in these subgroups likely reflects smaller sample sizes but may also indicate that the strictly unidimensional latent-factor model captures the symptom structure less precisely in these populations. Future studies with larger postmenopausal and male cohorts should investigate whether multi-factor solutions provide better fit and reveal additional substructure—such as a secondary immuno-inflammatory or PEM-related subdimension in women classified as postmenopausal or a more centralized autonomic-respiratory pole in men. Likewise, the four-factor solution that emerged in the exploratory factor analysis of male patients (compared with the one-factor solution in women) suggests that the symptom architecture may be genuinely more compartmentalized in male patients, although this finding should be interpreted with caution given the limited male sample size.

An additional limitation concerns the interpretation of the identified physiological dysregulation factor. The discussion proposes that this factor may reflect a chronic PEM-associated illness or danger response. However, this interpretation represents a conceptual framework derived from observed symptom associations rather than a directly tested hypothesis. The present dataset does not include longitudinal assessments of PEM episodes or biological measures of immune activation, autonomic function, vascular regulation, or metabolic processes. Consequently, the proposed framework cannot be directly validated using the current data. The findings therefore do not allow conclusions regarding whether the latent factor reflects a single underlying mechanism or the convergence of several interacting pathophysiological processes. Future studies combining longitudinal symptom assessments with physiological and molecular biomarkers will be required to evaluate these hypotheses.

Finally, although the analyses were repeated in a stratified training dataset to assess robustness and yielded directionally consistent results, true external validation in an independent ME/CFS cohort remains essential. Replication in cohorts using different diagnostic criteria, recruitment strategies, or geographic settings would substantially strengthen the generalizability of the identified physiological dysregulation factor and its sex- and menopause-specific expressions.

10. Conclusion

In conclusion, this study identifies a stable physiological dysregulation cluster in ME/CFS that links muscle symptoms with respiratory, cardiovascular, thermoregulatory, visual, flu-like, and gastrointestinal manifestations. Across multiple analytical approaches, muscle symptoms emerged as a central bridge symptom connecting several physiological domains, supporting the view that they reflect a broader multisystem process rather than an isolated disorder of skeletal muscle.

The observed symptom architecture is consistent with the existence of a common underlying physiological factor that remains stable across disease duration but differs in its clinical expression according to sex and menopausal status. Women classified as premenopausal were characterized primarily by cardiovascular, visual, and thermoregulatory symptom associations, whereas women classified as postmenopausal and men showed greater prominence of breathing-related symptoms. The persistence of flu-like symptoms in women classified as postmenopausal suggests that this subgroup retains distinctive features despite partial convergence toward the male phenotype.

Beyond autonomic, vascular, immunological, and metabolic interpretations, our findings are also compatible with the hypothesis that the identified physiological dysregulation factor represents the symptomatic manifestation of a chronic PEM-associated illness response. Within such a framework, muscle symptoms may constitute a particularly visible expression of a system-wide response involving multiple organ systems and physiological pathways.

Taken together, these findings encourage the inclusion of sex and menopausal status in future ME/CFS phenotyping strategies and highlight the importance of investigating physiological dysregulation as an integrated, multisystem phenomenon. Future longitudinal and mechanistic studies will be required to determine how the identified symptom architecture relates to PEM dynamics and to the biological pathways underlying disease heterogeneity in ME/CFS.

Declarations

1. Funding

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2. Conflicts of interest

The corresponding author declares on behalf of all authors that there is no conflict of interest.

3. Ethic declaration

This study was conducted in 2022 in accordance with the Declaration of Helsinki and approved by the Ethics Committee of Furtwangen University, Germany (application number: 22-057).

4. Consent to participate & Written Consent for publication

All participants were fully informed about the objectives and procedure before the study began and gave their written consent to participate. See *Information and declaration of consent* in the "Supplementary files" section

5. Consent for publication

Not applicable

6. Availability of data and material

All data supporting the results of this study are included in the article or can be requested from the corresponding author.

7. Code availability

Not applicable.

8. Author's Contributions

(1) LHH contributed to the development and conception of the research project, the preparation, collection, procurement, and provision of the data, the analysis and interpretation of the data, the review of the sources, the conclusions drawn from these findings, and the writing of the manuscript.
(2) LMH contributed to the preparation, provision, and interpretation of the data.

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Additional Notes

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